

LAB EXPERIMENT RECORD

Set 15: Product Sales Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze the first-quarter product sales data using a grouped bar chart, stacked area chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of Quarterly Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

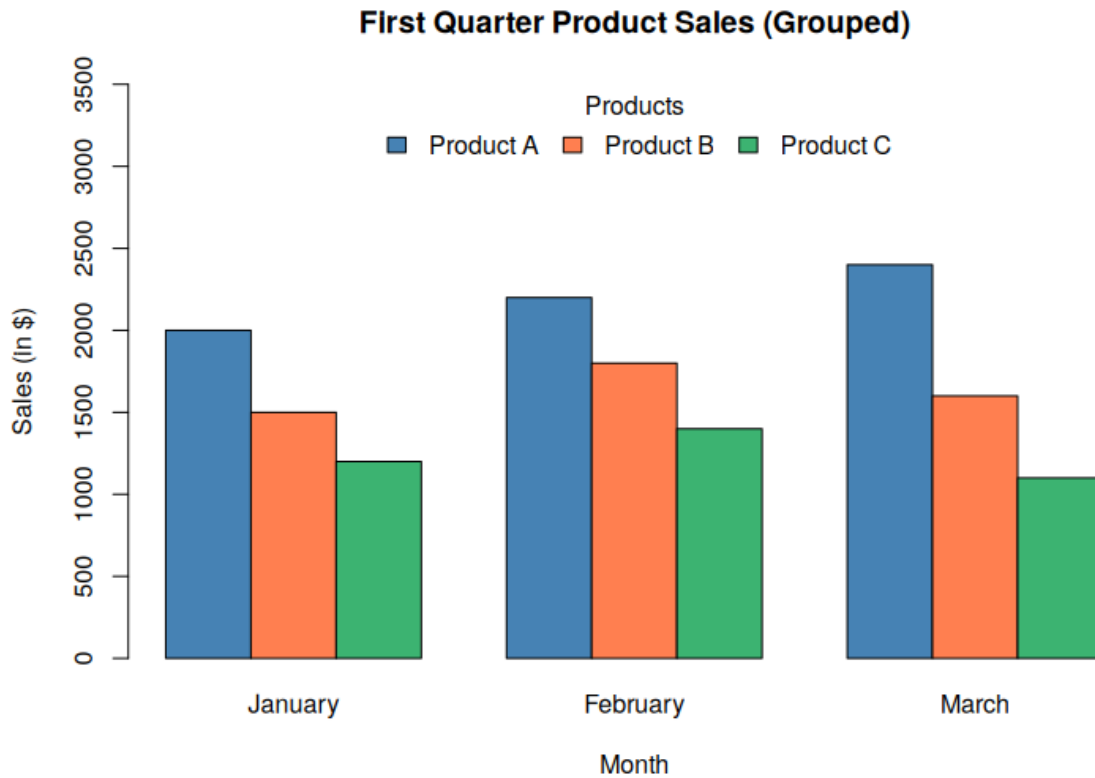
```
# Create dataset for Set 15
product_sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)

print(product_sales_data)

# Convert sales columns into a matrix for grouped plotting
sales_matrix <- as.matrix(product_sales_data[, c("January_Sales", "February_Sales",
"March_Sales")])
rownames(sales_matrix) <- product_sales_data$Product_Name

# 1. Grouped Bar Chart
barplot(sales_matrix,
  beside = TRUE,
  col = c("steelblue", "coral", "mediumseagreen"),
  xlab = "Month",
  ylab = "Sales (in $)",
```

```
main = "First Quarter Product Sales (Grouped)",
names.arg = c("January", "February", "March"),
ylim = c(0, 3500),
legend.text = rownames(sales_matrix),
args.legend = list(title = "Products", x = "top", horiz = TRUE, bty = "n"))
```



Output 1: First Quarter Product Sales (Grouped)

Observation: Product A has the highest sales in every month and grows steadily from 2000 to 2400. Product B peaks in February (1800), while Product C falls to its lowest value of 1100 in March.

PROGRAM 2: Stacked Area Chart of Overall Sales Trend

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
# Create dataset for Set 15
product_sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)

print(product_sales_data)

# Convert sales columns into a matrix
sales_matrix <- as.matrix(product_sales_data[, c("January_Sales", "February_Sales",
"March_Sales")])
rownames(sales_matrix) <- product_sales_data$Product_Name

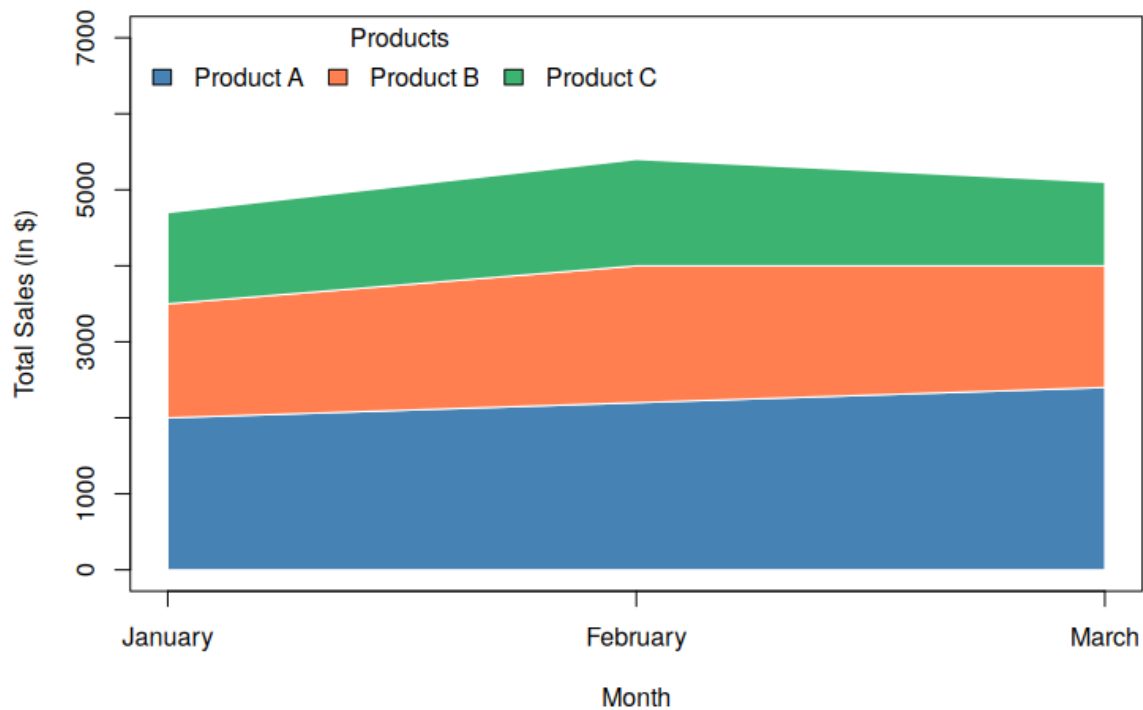
# Cumulative totals for stacking
month_no <- 1:3
cum_sales <- apply(sales_matrix, 2, cumsum)
area_cols <- c("steelblue", "coral", "mediumseagreen")

# 2. Stacked Area Chart for Overall Sales Trend
plot(month_no, cum_sales[3, ], type = "n",
      xaxt = "n",
      ylim = c(0, 7000),
      xlab = "Month",
      ylab = "Total Sales (in $)",
      main = "Overall Product Sales Trend (Stacked Area)")
axis(1, at = month_no, labels = c("January", "February", "March"))

lower <- rep(0, 3)
for (i in 1:nrow(sales_matrix)) {
  upper <- cum_sales[i, ]
  polygon(c(month_no, rev(month_no)), c(upper, rev(lower)),
          col = area_cols[i], border = "white")
  lower <- upper
}

legend("topleft",
      legend = rownames(sales_matrix),
      fill = area_cols,
      title = "Products",
      horiz = TRUE, bty = "n")
```

Overall Product Sales Trend (Stacked Area)



Output 2: Overall Product Sales Trend (Stacked Area)

Observation: Total sales rise from 4700 in January to the highest value of 5400 in February and then fall slightly to 5100 in March. Product A contributes the largest share of total sales in every month.

PROGRAM 3: Table of Monthly Sales Data

Build a table to show the monthly sales data for each product. Label the table elements.

```
# Create dataset for Set 15
product_sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)

print(product_sales_data)

# 3. Table of monthly sales data with labelled columns
formatted_table <- product_sales_data
colnames(formatted_table) <- c("Product ID", "Product Name",
                              "Jan Sales ($)", "Feb Sales ($)", "Mar Sales ($)")

cat("=== MONTHLY PRODUCT SALES TABLE ===\n")
print(formatted_table, row.names = FALSE)
```

```
=== MONTHLY PRODUCT SALES TABLE ===
> print(formatted_table, row.names = FALSE)
  Product ID Product Name Jan Sales ($) Feb Sales ($) Mar Sales ($)
        1    Product A      2000      2200      2400
        2    Product B      1500      1800      1600
        3    Product C      1200      1400      1100
>
```

Output 3: Monthly Product Sales Table

Observation: The table lists the January, February, and March sales of each product with clearly labelled columns, making it easy to read and compare the exact values.

PROGRAM 4: Interactive Dashboard for Product Sales

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 15
product_sales_data <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
prod_cols <- c("steelblue", "coral", "mediumseagreen")
names(prod_cols) <- product_sales_data$Product_Name

ui <- fluidPage(
  titlePanel("Product Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("prod", "Product:",
        choices = names(prod_cols), selected = names(prod_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("groupPlot", height = "300px")),
        column(6, plotOutput("areaPlot", height = "300px"))
      ),
      h4("Monthly Product Sales Table"),
      tableOutput("salesTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(product_sales_data, Product_Name %in% input$prod) })

  output$groupPlot <- renderPlot({
    d <- filtered()
    m <- as.matrix(d[, 2:4]); rownames(m) <- d$Product_Name
    barplot(m, beside = TRUE, col = prod_cols[d$Product_Name], ylim = c(0, 3500),
      names.arg = c("January", "February", "March"), xlab = "Month",
      ylab = "Sales (in $)", main = "First Quarter Sales (Grouped)",
      legend.text = rownames(m),
      args.legend = list(x = "top", horiz = TRUE, bty = "n", cex = 0.8))
  }, width = 420, height = 300, res = 96)

  output$areaPlot <- renderPlot({
    d <- filtered()
    m <- as.matrix(d[, 2:4])
    cum <- apply(m, 2, cumsum); if (is.null(dim(cum))) cum <- matrix(cum, nrow = 1)
    plot(1:3, cum[nrow(cum), ], type = "n", xaxt = "n", ylim = c(0, 7000),
```

```

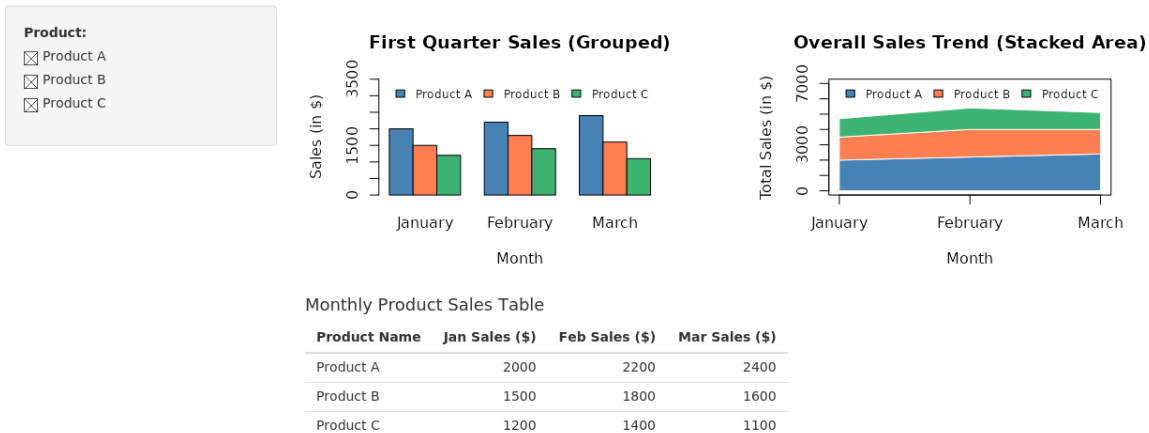
      xlab = "Month", ylab = "Total Sales (in $)", main = "Overall Sales Trend (Stacked
Area)")
      axis(1, at = 1:3, labels = c("January", "February", "March"))
      lower <- rep(0, 3)
      for (i in seq_len(nrow(cum))) {
        polygon(c(1:3, 3:1), c(cum[i, ], rev(lower)), col = prod_cols[d$Product_Name[i]], border =
"white")
        lower <- cum[i, ]
      }
      legend("top", legend = d$Product_Name, fill = prod_cols[d$Product_Name],
        horiz = TRUE, bty = "n", cex = 0.8)
    }, width = 420, height = 300, res = 96)

output$salesTable <- renderTable({
  d <- filtered()
  colnames(d) <- c("Product Name", "Jan Sales ($)", "Feb Sales ($)", "Mar Sales ($)")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Product Sales Dashboard



Output 4: Product Sales Interactive Dashboard (R Shiny)

Observation: The dashboard shows the grouped bar chart, the stacked area chart, and the sales table together. Unticking a product in the Product filter removes it from both charts and the table at once, so the sales of selected products can be explored together.

Procedure

1. Create the product sales data frame with January, February, and March sales.
2. Convert the sales columns into a matrix and use `barplot()` with `beside = TRUE` to draw the grouped bar chart.
3. Calculate cumulative sales with `apply()` and `cumsum()` and draw the stacked area chart using `polygon()`.
4. Rename the columns with `colnames()` and print the labelled table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The monthly product sales data was successfully visualized using a grouped bar chart, stacked area chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>as.matrix()</code>	Converts sales columns into a matrix for plotting.
<code>barplot()</code>	Creates the grouped bar chart (<code>beside = TRUE</code>).
<code>apply()</code> / <code>cumsum()</code>	Calculates cumulative sales for stacking.
<code>polygon()</code>	Draws the filled layers of the stacked area chart.
<code>colnames()</code>	Sets labelled column names for the table.
<code>print()</code>	Displays the formatted sales table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.